

Markus Frohmann

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Education

Johannes Kepler University

Linz, Austria

M.S. in Artificial Intelligence

Oct 2023 – expected Apr 2026

- GPA: 1.25 (Austrian scale: 1.0-5.0, 1.0 is best; equivalent to $\approx 3.75/4.0$)
- Thesis: *Robust AI-Generated Lyrics Detection* (Advisor: Prof. Markus Schedl)
- Visiting Student, *National Taiwan University of Science and Technology*, Taiwan (Spring 2025)

Johannes Kepler University

Linz, Austria

B.S. in Artificial Intelligence, with Distinction

Oct 2020 – Sep 2023

- GPA: 1.2 (equivalent to $\approx 3.8/4.0$)
- Thesis: *Simple and Efficient Task Transfer by Learning to Scale*. (Advisor: Dr. Navid Rekasaz)

Research Experience

Deezer Research

Paris, France

Supervisors: Dr. Elena Epure, Dr. Gabriel Meseguer-Brocal

Sep 2024 – Feb 2025

- Designed methods to robustly detect synthetic (AI-generated) lyrics from audio and text.

ETH Zürich, Institute for Machine Learning (Rycolab)

Zürich, Switzerland

Supervisors: Prof. Ryan Cotterell, Dr. Mario Giulianelli

Jul 2024 – Sep 2024

- Introduced frameworks to build any-to-any multimodal models, focusing on video understanding.

Johannes Kepler University, Institute of Computational Perception

Linz, Austria

Advisors: Prof. Markus Schedl, Dr. Navid Rekasaz

Mar 2023 – Jul 2024

- Led research on various modular, efficient, and multilingual methods in NLP.

Professional Experience

Bosch

Linz, Austria

Data Scientist, Powertrain Solutions

Mar 2023 – Dec 2023

- Developed production-grade predictive maintenance models and created a company-wide MLOps curriculum.

Leftshift One

Graz, Austria

Machine Learning Engineer

Jul 2021 – Oct 2022

- Designed, built and deployed end-to-end machine learning solutions for various customers.

Publications

CONFERENCE PAPERS

Double Entendre: Robust Audio-Based AI-Generated Lyrics Detection via Multi-View Fusion

M Frohmann, G Meseguer-Brocal, M Schedl, E V Epure
ACL 2025 (Findings)

AI-Generated Song Detection via Lyrics Transcripts

M Frohmann, E V Epure, G Meseguer-Brocal, M Schedl, R Hennequin
ISMIR 2025

Segment Any Text: A Universal Approach for Robust, Efficient and Adaptable Sentence Segmentation

M Frohmann, I Sterner, I Vulić, B Minixhofer, M Schedl
EMNLP 2024 (Main Conference)

Unlabeled Debiasing in Downstream Tasks via Class-wise Low Variance Regularization

S Masoudian, M Frohmann, N Rekasaz, M Schedl
EMNLP 2024 (Main Conference)

ScaLearn: Simple and Highly Parameter-Efficient Task Transfer by Learning to Scale

M Frohmann, C Holtermann, S Masoudian, A Lauscher, N Rekabsaz

ACL 2024 (Findings)

What the Weight?! A Unified Framework for Zero-Shot Knowledge Composition

C Holtermann, **M Frohmann**, N Rekabsaz, A Lauscher

EACL 2024 (Findings)

Fine-Grained Sentiment Control via Lightweight Modular Rewriting

S Masoudian, **M Frohmann**, N Rekabsaz, M Schedl Manuscript in preparation, 2026

Punctuation Restoration as a Language-Agnostic Pretraining Objective

C Min, **M Frohmann**, F Rex, E S Lee, E G Wilcox Manuscript in preparation, 2026

JOURNAL ARTICLES

Predicting the Price of Bitcoin Using Sentiment-Enriched Time Series Forecasting

M Frohmann, M Karner, S Khudoyan, R Wagner, M Schedl

Big Data and Cognitive Computing 7(3):137 (2023)

WORKSHOP PAPERS

Synthetic Lyrics Detection Across Languages and Genres

Y Labrak, **M Frohmann**, G Meseguer-Brocal, E V Epure

5th Workshop on Trustworthy NLP (TrustNLP 2025)

jkulinzstudents Submission to the MARIO Challenge 2024: Task 1

M Huber, P Binder, **M Frohmann**

MICCAI 2024 Workshops, LNCS 15503 (Springer)

jkulinzstudents Submission to the MARIO Challenge 2024: Task 2

P Binder, M Huber, **M Frohmann**

MICCAI 2024 Workshops, LNCS 15503 (Springer)

SOFTWARE & OPEN SOURCE CONTRIBUTIONS

wtpsplit

Main Contributor

 [Repository](#)

 [PyPI](#)

- Open-source text segmentation library from the EMNLP 2024 paper *Segment Any Text*; **1.2k+ GitHub stars**.

Honors & Awards

- 2025 **Recipient**, Taiwan-Europe Connectivity Scholarship, *NTUST* (EU-Taiwan Academic Exchange)
- 2020 – 2023 **Recipient**, Yearly Merit Scholarship, *Johannes Kepler University*
- 2022 **Winner**, Student Challenge on Music Recommendation, *Johannes Kepler University*
- 2021 **Winner**, Student Team Challenge on Music Emotion Classification, *Johannes Kepler University*

Mentoring & Supervision

M.S. Thesis Co-Supervisor, Johannes Kepler University

- Guiding an M.S. student in developing advanced chunking strategies for Retrieval-Augmented Generation (RAG).

Research Mentor, High School Project

- Mentored a student through the complete research cycle on analyzing next-generation energy storage solutions.

Academic Activities

- 2025 **Peer Reviewer**, ACL Rolling Review (ARR)
- 2024–2025 **Conference Presentations**, Posters – ACL '24, EMNLP '24; Oral – ISMIR '25

Technical Skills & Languages

Programming Python, SQL, Linux/Bash

Frameworks PyTorch, Transformers, Pandas, NumPy, Scikit-learn, PySpark

Tools Git, Docker, MLFlow, Weights & Biases, Hadoop, ONNX

Human German (Native), English (Full Professional), French (Working), Mandarin (Intermediate)